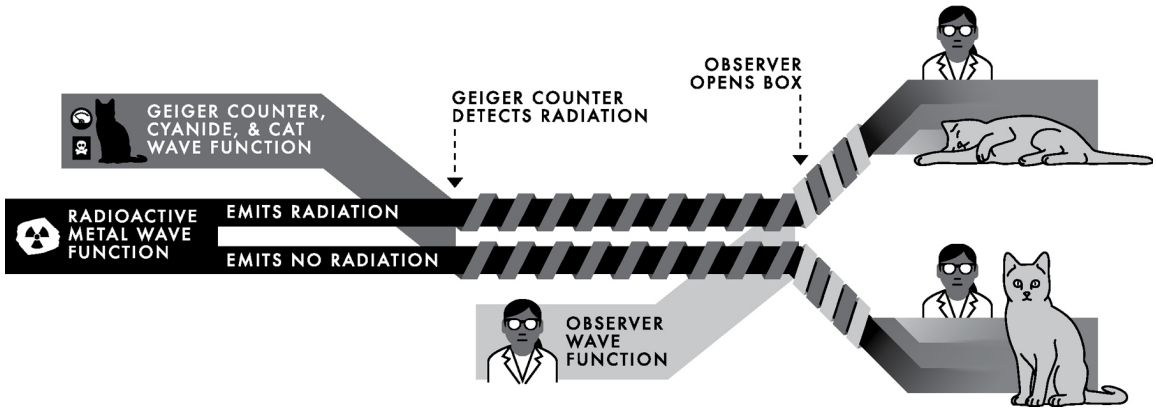


Figure 6.2. Branching in the many-worlds interpretation.

Of course, it doesn't seem like we split in two when we perform experiments—or anytime at all for that matter. But Everett had an answer ready for this as well. If I ask the "you" that sees the living cat how many cats you see, you'll answer "just one." And if I ask the same question of the "you" in the other branch of the wave function, the one with the dead cat, the answer will be the same (though your tone of voice will probably be quite different). The same thing happens, Everett pointed out, if I ask each copy of you how many selves you see. There is only one copy of you in each branch of the wave function, and, even if you repeat the experiment, this will still be true—there will be more branches, but each branch will still only have one copy of you. And the Schrödinger equation dictates that each branch will carry on independently of the others, with hardly any interaction between branches.

Strange enough, but we're not done. As you interact with the things in your environment, they get entangled with you, and then other things get entangled with them, and so on. Eventually, we have a single complicated and messy wave function for the entire universe—the universal wave function. And as more events happen, that universal wave function splits into more and more noninteracting parts, each merrily marching along to the deterministic beat of the Schrödinger equation. These are the many worlds of Everett's interpretation. They may seem absurd on the face of it: there is, after all, only one world that we experience. But if that's your objection, Everett's reply is that you're hardly alone: to each person in each branch of the universal wave function, their world appears to be the only